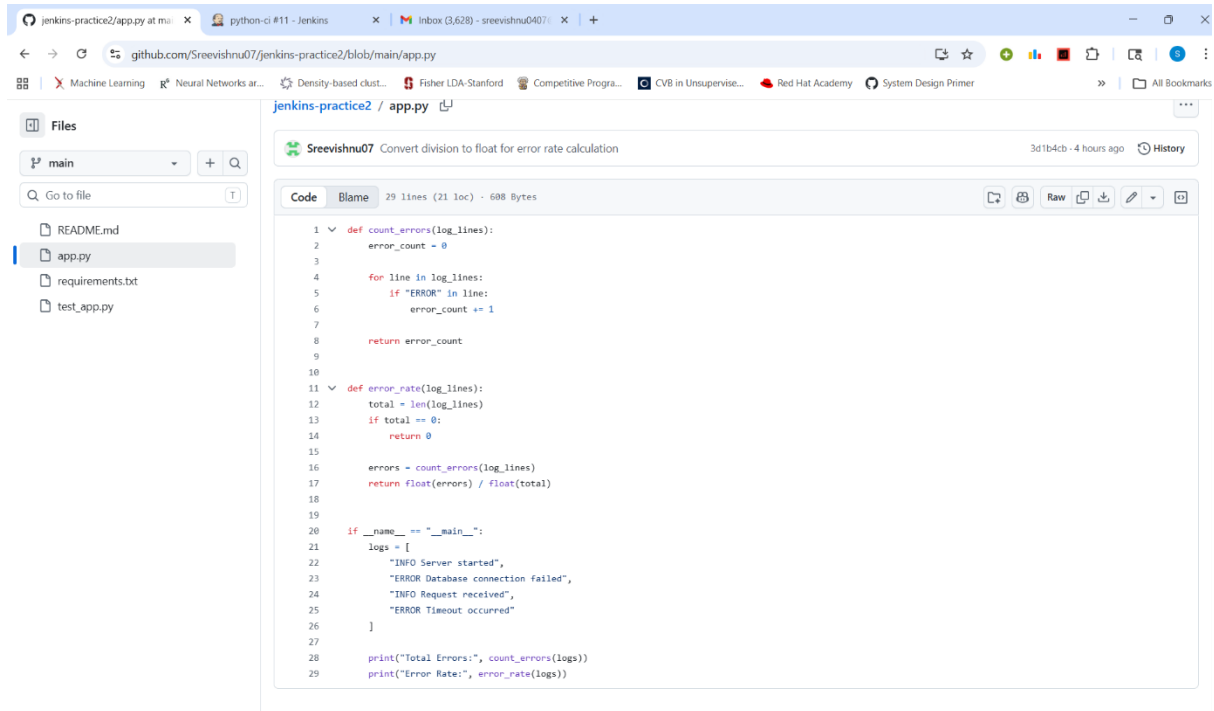


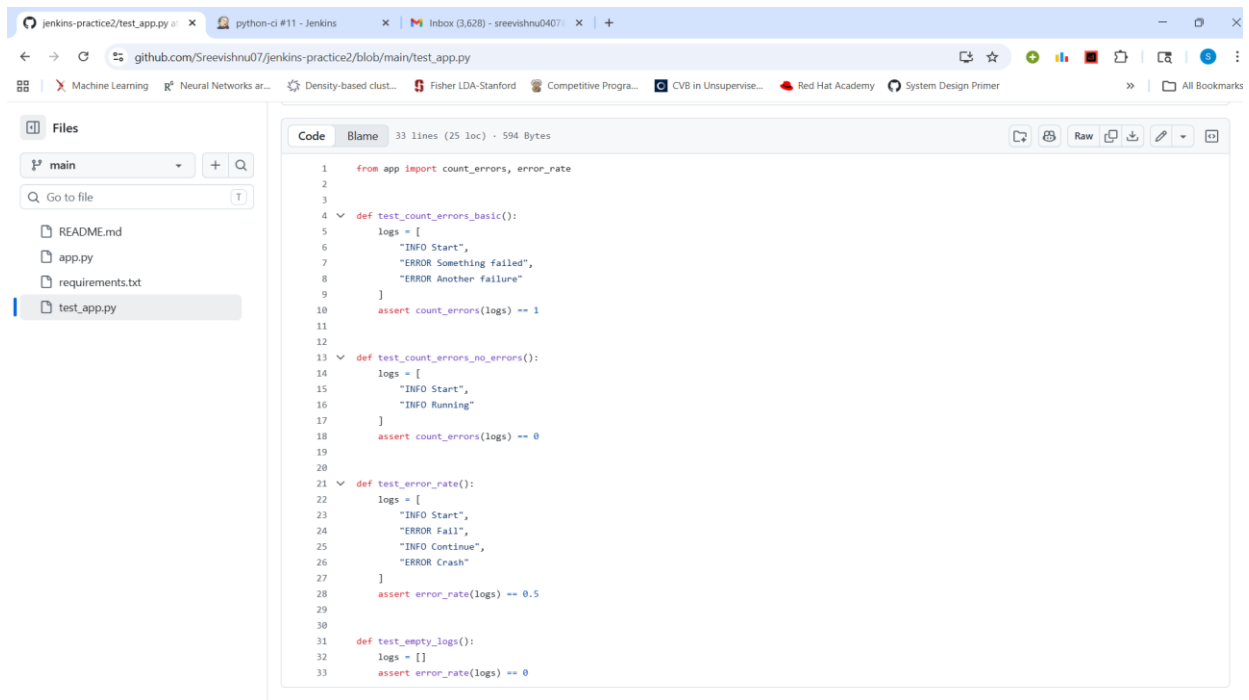
Special practicals- Jenkins (Final Armageddon) Part -1

1. Jenkins-Based Python CI Pipeline with Webhook Automation and Email Notifications



The screenshot shows a web browser displaying a GitHub repository named 'jenkins-practice2' by user 'Sreevishnu07'. The file 'app.py' is selected, showing 29 lines of Python code. The code defines functions for counting errors and calculating error rates, and includes a main block for logging and printing results.

```
1 def count_errors(log_lines):
2     error_count = 0
3
4     for line in log_lines:
5         if "ERROR" in line:
6             error_count += 1
7
8     return error_count
9
10
11 def error_rate(log_lines):
12     total = len(log_lines)
13     if total == 0:
14         return 0
15
16     errors = count_errors(log_lines)
17     return float(errors) / float(total)
18
19
20 if __name__ == "__main__":
21     logs = [
22         "INFO Server started",
23         "ERROR Database connection failed",
24         "INFO Request received",
25         "ERROR Timeout occurred"
26     ]
27
28     print("Total Errors:", count_errors(logs))
29     print("Error Rate:", error_rate(logs))
```



The screenshot shows a web browser displaying the same GitHub repository, but with the file 'test_app.py' selected. The file contains 33 lines of Python code, which are unit tests for the 'app.py' file. The tests verify the 'count_errors' and 'error_rate' functions using 'assert' statements.

```
1 from app import count_errors, error_rate
2
3
4 def test_count_errors_basic():
5     logs = [
6         "INFO Start",
7         "ERROR Something failed",
8         "ERROR Another failure"
9     ]
10     assert count_errors(logs) == 1
11
12
13 def test_count_errors_no_errors():
14     logs = [
15         "INFO Start",
16         "INFO Running"
17     ]
18     assert count_errors(logs) == 0
19
20
21 def test_error_rate():
22     logs = [
23         "INFO Start",
24         "ERROR Fail",
25         "INFO Continue",
26         "ERROR Crash"
27     ]
28     assert error_rate(logs) == 0.5
29
30
31 def test_empty_logs():
32     logs = []
33     assert error_rate(logs) == 0
```

```
D:\downloads\ngrok-v3-stabl x + v
ngrok (Ctrl+C to quit)

Take our ngrok in production survey! https://forms.gle/aXiBFWzEA36DudFn6

Session Status      online
Account             sreevishnu0407@gmail.com (Plan: Free)
Update              update available (version 3.39.1, Ctrl-U to update)
Version             3.37.6
Region              Europe (eu)
Latency              204ms
Latency             78ms
Web Interface        http://127.0.0.1:4040
Forwarding           https://cartridge-gooey-ninth.ngrok-free.dev -> http://localhost:8080

Connections          ttl    opn    rt1    rt5    p50    p90
                    4      0      0.00   0.00   30.28   31.32

HTTP Requests
-----
22:57:19.251 IST POST /github-webhook/      200 OK
22:54:16.722 IST POST /github-webhook/      200 OK
19:08:01.204 IST POST /github-webhook/      200 OK
19:07:27.121 IST POST /github-webhook/      200 OK
```

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
1 pipeline {
2   agent any
3   stages{
4     stage('Clone Repo'){
5       steps{
6         git branch: 'main',url: 'https://github.com/Sreevishnu07/jenkins-practice2'
7       }
8     }
9     stage('Install dependencies'){
10      steps{
11        bat 'python -m pip install -r requirements.txt'
12      }
13    }
14    stage('Run tests'){
15      steps{
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Definition

Pipeline script

Script ?

```
16         bat 'python -m pytest'
17     }
18 }
19 }
20 post{
21     success {
22         echo 'All tests passed'
23     }
24     failure {
25         emailxnt (
26             subject: "Build failed: ${env.JOB_NAME}",
27             body: "Check logs: ${env.BUILD_URL}",
28             to: "sreevishnu0407@gmail.com"
29         )
30     }
}
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

The screenshot shows a web browser window with multiple tabs. The active tab is 'python-ci #9 - Jenkins'. The address bar shows 'localhost:8080/job/python-ci/9/'. The browser's bookmark bar contains various links like 'Machine Learning', 'Neural Networks ar...', 'Density-based clust...', 'Fisher LDA-Stanford', 'Competitive Progra...', 'CVB in Unsupervise...', 'Red Hat Academy', and 'System Design Primer'. The Jenkins interface shows the build status as 'Success' with a green checkmark. The build is labeled '#9 (May 5, 2026, 7:08:08 PM)'. On the left, there is a sidebar with navigation links: Status, Changes, Console Output, Edit Build Information, Delete build '#9', Polling Log, Timings, Git Build Data, Pipeline Overview, Restart from Stage, Replay, Pipeline Steps, Workspaces, Previous Build, and Next Build. The main content area shows the build details: 'Started by GitHub push by Sreevishnu07', 'This run spent: 5.1 sec waiting; 9.3 sec build duration; 14 sec total from scheduled to completion.', 'Revision: 6ea336d481b61e67cb3d8728486398288b7fef94', 'Repository: https://github.com/Sreevishnu07/jenkins-practice2', and 'refs/remotes/origin/main'. Below this, there are two commit entries: 'Create README.md for Jenkins Python CI pipeline' and 'trigger for webhook checking', both with timestamps and commit hashes.

Jenkins / python-ci / #9

Status

Changes

Console Output

Edit Build Information

Delete build '#9'

Polling Log

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

Next Build

#9 (May 5, 2026, 7:08:08 PM)

Add description

Keep this build forever

Started 3 hr 54 min ago

Took 9.3 sec

Started by GitHub push by Sreevishnu07

This run spent:

- 5.1 sec waiting;
- 9.3 sec build duration;
- 14 sec total from scheduled to completion.

Revision: 6ea336d481b61e67cb3d8728486398288b7fef94

Repository: <https://github.com/Sreevishnu07/jenkins-practice2>

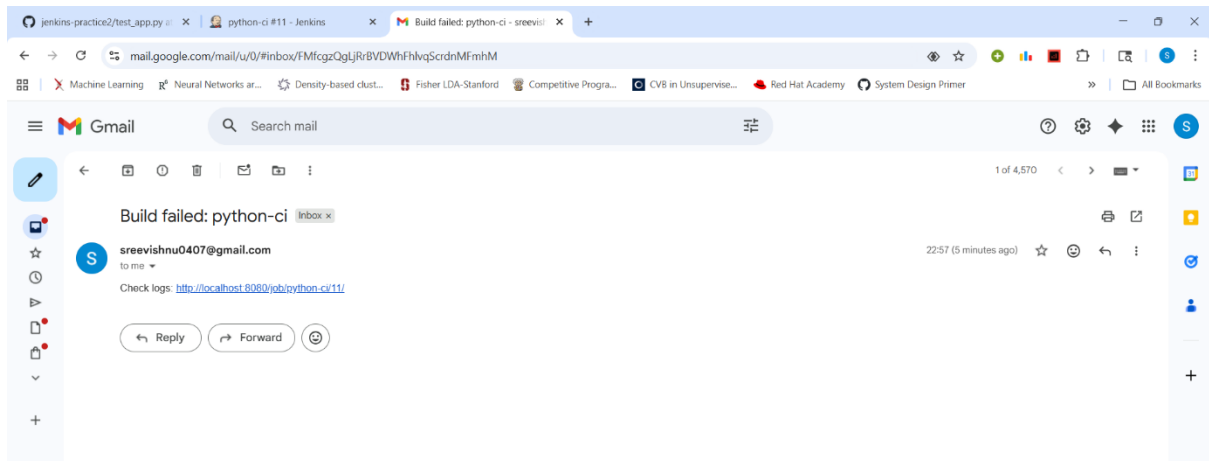
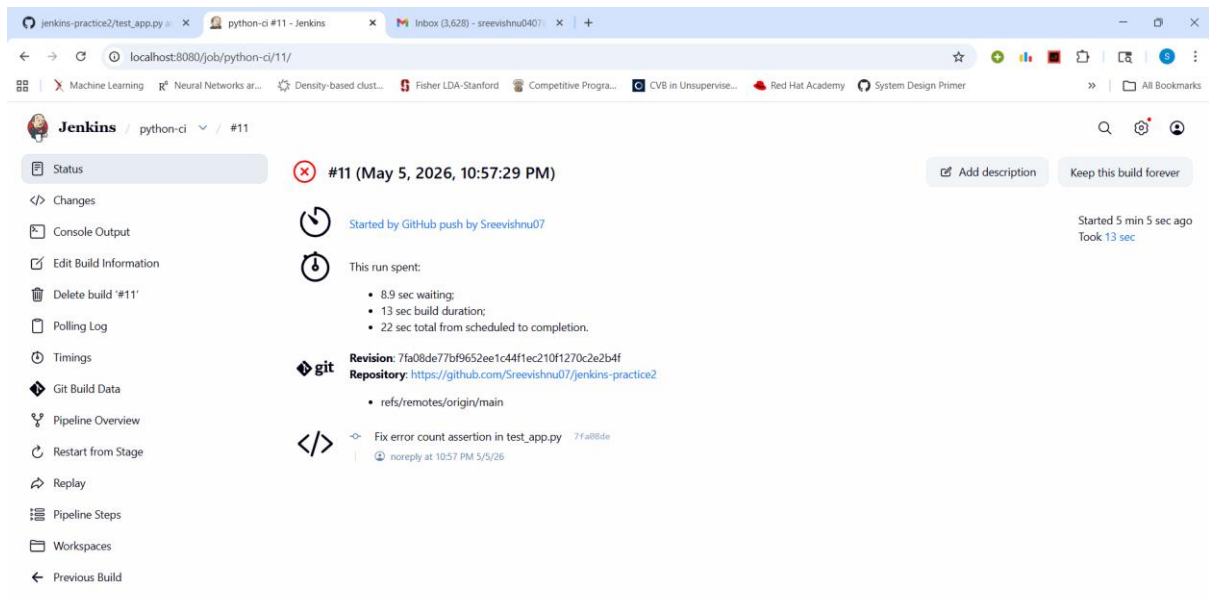
- refs/remotes/origin/main

Create README.md for Jenkins Python CI pipeline 274e5ee

noreply at 7:03 PM 5/5/26

trigger for webhook checking 6ea336d

noreply at 7:07 PM 5/5/26



Above 2 pics depict - On failure, Email is sent to the owner

2. Design and Implementation of a CI Pipeline using Jenkins, Maven, and Docker for Java Applications (done via both Pipeline script and Jenkinsfile)

Sreevishnu07/jenkins-practice3

github.com/Sreevishnu07/jenkins-practice3

Machine LearningNeural Networks ar...Density-based clust...Fisher LDA-StanfordCompetitive Progra...CVB in Unsupervise...Red Hat AcademySystem Design PrimerAll Bookmark

Sreevishnu07 / jenkins-practice3

Type to search

CodeIssuesPull requestsAgentsActionsProjectsWikiSecurity and qualityInsightsSettings

jenkins-practice3Public

PinWatch 0Fork 0Star 0

main1 Branch0 Tags

Go to file

Add fileCode

Sreevishnu07 Update JUnit version to 4.13.24505dc1 · 1 minute ago5 Commits

srcadded proj files26 minutes ago

Dockerfileadded proj files26 minutes ago

README.mdFix typo in README CI workflow description9 minutes ago

pom.xmlUpdate JUnit version to 4.13.21 minute ago

README

Docker + Maven + Jenkins CI Pipeline

- Built a Java application using Maven with standard project structure and unit tests (JUnit)
- Implemented Continuous Integration using Jenkins Pipeline (Jenkinsfile)
- Automated build process using `mvn clean install (compile + test)`
- Containerized the application using Docker for consistent runtime environment
- Docker image builds include project compilation and test execution
- Integrated GitHub with Jenkins using webhooks for automatic pipeline triggering on push

About

End-to-end CI pipeline using Jenkins, Maven, and Docker for a Java application with automated testing, containerization, and webhook-based build triggers.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 1

jenkins-practice3/src/main/java

github.com/Sreevishnu07/jenkins-practice3/blob/main/src/main/java/com/example/App.java

Machine LearningNeural Networks ar...Density-based clust...Fisher LDA-StanfordCompetitive Progra...CVB in Unsupervise...Red Hat AcademySystem Design PrimerAll Bookmark

Sreevishnu07 / jenkins-practice3

Type to search

CodeIssuesPull requestsAgentsActionsProjectsWikiSecurity and qualityInsightsSettings

Files

main

Go to file

src

main/java/com/example

App.java

test/java/com/example

Dockerfile

README.md

pom.xml

jenkins-practice3 / src / main / java / com / example / App.java

Sreevishnu07 added proj filesa647960 · 36 minutes agoHistory

CodeBlame

20 lines (18 loc) · 401 Bytes

CopyRawDownloadEdit

```
1 package com.example;
2
3 public class App
4 {
5     public static int add(int a,int b){
6         return a+b;
7     }
8
9     public static int div(int a,int b){
10        if(b==0){
11            throw new IllegalArgumentException("You cannot divide by zero");
12        }
13        return a/b;
14    }
15    public static void main( String[] args )
16    {
17        System.out.println( "Sum: "+add(2,3) );
18        System.out.println("Div: "+div(6,2) );
19    }
20 }
```

Symbols

Find definitions and references for functions and other symbols in this file by clicking a symbol below or in the code.

Filter symbols

class App

func add

func div

func main

jenkins-practice3/src/test/java/

github.com/Sreevishnu07/jenkins-practice3/blob/main/src/test/java/com/example/AppTest.java

Sreevishnu07 / jenkins-practice3

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Files

main

Go to file

src

main/java/com/example

App.java

test/java/com/example

AppTest.java

Dockerfile

README.md

pom.xml

jenkins-practice3 / src / test / java / com / example / AppTest.java

Sreevishnu07 added proj files a647960 - 36 minutes ago History

Code Blame 23 lines (19 loc) - 411 Bytes

```
1 package com.example;
2
3 import org.junit.Test;
4 import static org.junit.Assert.*;
5
6 public class AppTest{
7     @Test
8     public void testAdd(){
9         int res=App.add(2,3);
10        assertEquals(5,res);
11    }
12
13    @Test
14    public void testdiv(){
15        int res=App.div(6,2);
16        assertEquals(3,res);
17    }
18
19    @Test(expected = IllegalArgumentException.class)
20    public void testDivideByZero(){
21        App.div(10,0);
22    }
23 }
```

Symbols

Find definitions and references for functions and other symbols in this file by clicking a symbol below or in the code.

Filter symbols

class AppTest

func testAdd

func testdiv

func testDivideByZero

jenkins-practice3/Dockerfile at

github.com/Sreevishnu07/jenkins-practice3/blob/main/Dockerfile

Sreevishnu07 / jenkins-practice3

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Files

main

Go to file

src

main/java/com/example

App.java

test/java/com/example

AppTest.java

Dockerfile

README.md

pom.xml

jenkins-practice3 / Dockerfile

Sreevishnu07 added proj files a647960 - 37 minutes ago History

Code Blame 5 lines (5 loc) - 164 Bytes

```
1 FROM maven:3.9.6-eclipse-temurin-17
2 WORKDIR /app
3 COPY . .
4 RUN mvn clean install
5 CMD ["java", "-cp", "target/docker-maven-jenkins-1.0-SNAPSHOT.jar", "com.example.App"]
```

Dockerfile for maven-java environment

jenkins-practice3/pom.xml at main · Sreevishnu07/jenkins-practice3 · GitHub

Files

- main
- src
 - main/java/com/example
 - App.java
 - test/java/com/example
 - AppTest.java
 - Dockerfile
 - README.md
 - pom.xml

jenkins-practice3 / pom.xml

Sreevishnu07 Update JUnit version to 4.13.2 4505dc1 · 13 minutes ago History

Code Blame 18 lines (18 loc) · 665 Bytes

```
1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
2 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
3 <modelVersion>4.0.0</modelVersion>
4 <groupId>com.example</groupId>
5 <artifactId>docker-maven-jenkins</artifactId>
6 <packaging>jar</packaging>
7 <version>1.0-SNAPSHOT</version>
8 <name>docker-maven-jenkins</name>
9 <url>http://maven.apache.org/</url>
10 <dependencies>
11 <dependency>
12 <groupId>junit</groupId>
13 <artifactId>junit</artifactId>
14 <version>4.13.2</version>
15 <scope>test</scope>
16 </dependency>
17 </dependencies>
18 </project>
```

maven-docker-ci - Jenkins

localhost:8080/job/maven-docker-ci/

Jenkins / maven-docker-ci

Status

Changes

Build Now

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax

GitHub Hook Log

maven-docker-ci

Integration of maven-jenkins-docker

Permalinks

- Last build (#3), 45 sec ago
- Last stable build (#3), 45 sec ago
- Last successful build (#3), 45 sec ago
- Last failed build (#2), 4 min 53 sec ago
- Last unsuccessful build (#2), 4 min 53 sec ago
- Last completed build (#3), 45 sec ago

Buils >

Filter

Today

- #3 4:48 PM
- #2 4:44 PM
- #1 4:42 PM

maven-docker-ci Config - Jenkins

localhost:8080/job/maven-docker-ci/configure

Jenkins / maven-docker-ci / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

☐ Pipeline speed/durability override ?

☐ Preserve stashes from completed builds ?

☐ This project is parameterized ?

☐ Throttle builds ?

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

☐ Build after other projects are built ?

☐ Build periodically ?

☒ GitHub hook trigger for GITScm polling ?

☐ Poll SCM ?

☐ Trigger builds remotely (e.g., from scripts) ?

Define your pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
1 pipeline{
2   agent any
3   stages{
4     stage('1.Clone Repo'){
5       steps{
6         git branch: 'main',url: 'https://github.com/Sreevishnu07/jenkins-practice3.git'
7       }
8     }
9     stage('2.Build docker image'){
10      steps{
11        bat 'docker build -t maven-jenkins-docker-demo .'
12      }
13    }
14    stage('3.Run container'){
15      steps{
```

☒ Use Groovy Sandbox ?

Definition

Pipeline script

Script ?

```
16         bat 'docker run --rm maven-jenkins-docker-demo'
17     }
18 }
19 }
20 post{
21     success{
22         echo 'Build and docker run successful'
23     }
24     failure{
25         emailtext(
26             subject: "Build failed: ${env.JOB_NAME}",
27             body: "Check logs to find: ${env.BUILD_URL}",
28             to: "sreevishnu0407@gmail.com"
29         )
30     }
}
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Save Apply

jenkins-practice3/pom.xml at maven-docker-ci Config - Jenkins Build failed: maven-docker-ci

mail.google.com/mail/u/0/#inbox/FMfegzOgljSzbCVQktwGLdjZmZccvKr

Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CV8 in Unsupervise... Red Hat Academy System Design Primer All Bookmarks

Gmail Search mail

1 of 4,575

Build failed: maven-docker-ci

sreevishnu0407@gmail.com to me 16:43 (20 minutes ago)

Check logs to find: <http://localhost:8080/job/maven-docker-ci/1/>

sreevishnu0407@gmail.com to me 16:45 (17 minutes ago)

Check logs to find: <http://localhost:8080/job/maven-docker-ci/2/>

Reply Forward

Email triggers for 2 failed runs

```
ngrok
( Ctrl+C to quit)

One gateway for every AI model. Available in early access *now*: https://ngrok.com/r/ai

Session Status online
Account sreevishnu0407@gmail.com (Plan: Free)
Update update available (version 3.39.1, Ctrl-U to update)
Version 3.37.6
Region Europe (eu)
Latency 241ms
Web Interface http://127.0.0.1:4040
Forwarding https://cartridge-gooey-ninth.ngrok-free.dev -> http://localhost:8080

Connections
  ttl    opn    rt1    rt5    p50    p90
    4      0    0.00   0.00   30.11  31.67

HTTP Requests
-----
16:48:12.685 IST POST /github-webhook/ 200 OK
16:48:30.866 IST POST /github-webhook/ 200 OK
16:39:28.101 IST POST /github-webhook/ 200 OK
16:38:50.353 IST POST /github-webhook/ 200 OK
```

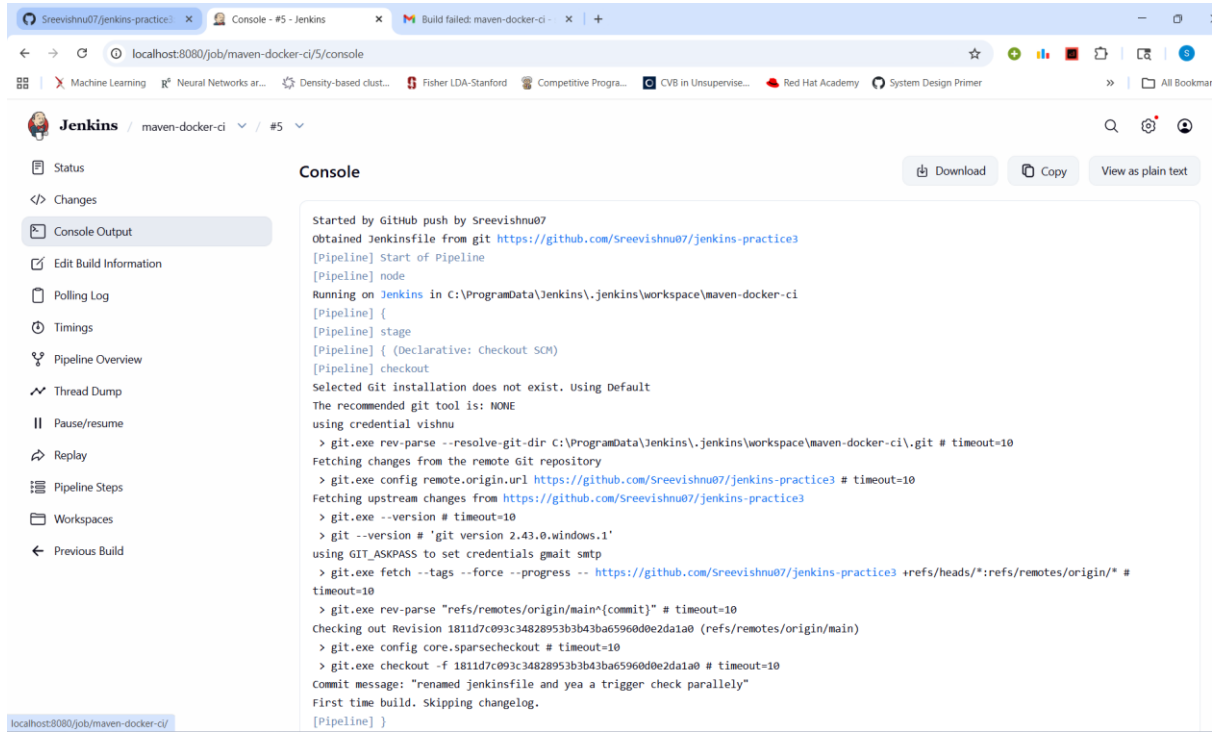
Ngrok for GitHub webhooks

```
jenkins-practice3/pom.xml at r... Console - #3 - Jenkins Build failed: maven-docker-ci - X +
localhost:8080/job/maven-docker-ci/3/console
Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CVB in Unsupervise... Red Hat Academy System Design Primer
Jenkins / maven-docker-ci #3
#9 exporting manifest list sha256:e00d5e080c40b9072/85199a04a0e394e7/4e6a/e9e143c90210b649f9e0d39 0.05 done
#9 naming to docker.io/library/maven-jenkins-docker-demo:latest done
#9 unpacking to docker.io/library/maven-jenkins-docker-demo:latest
#9 unpacking to docker.io/library/maven-jenkins-docker-demo:latest 0.4s done
#9 DONE 1.9s
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (3.Run container)
[Pipeline] bat

C:\ProgramData\Jenkins\jenkins\workspace\maven-docker-ci>docker run --rm maven-jenkins-docker-demo
Sum: 5
Div: 3
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Build and docker run successful
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

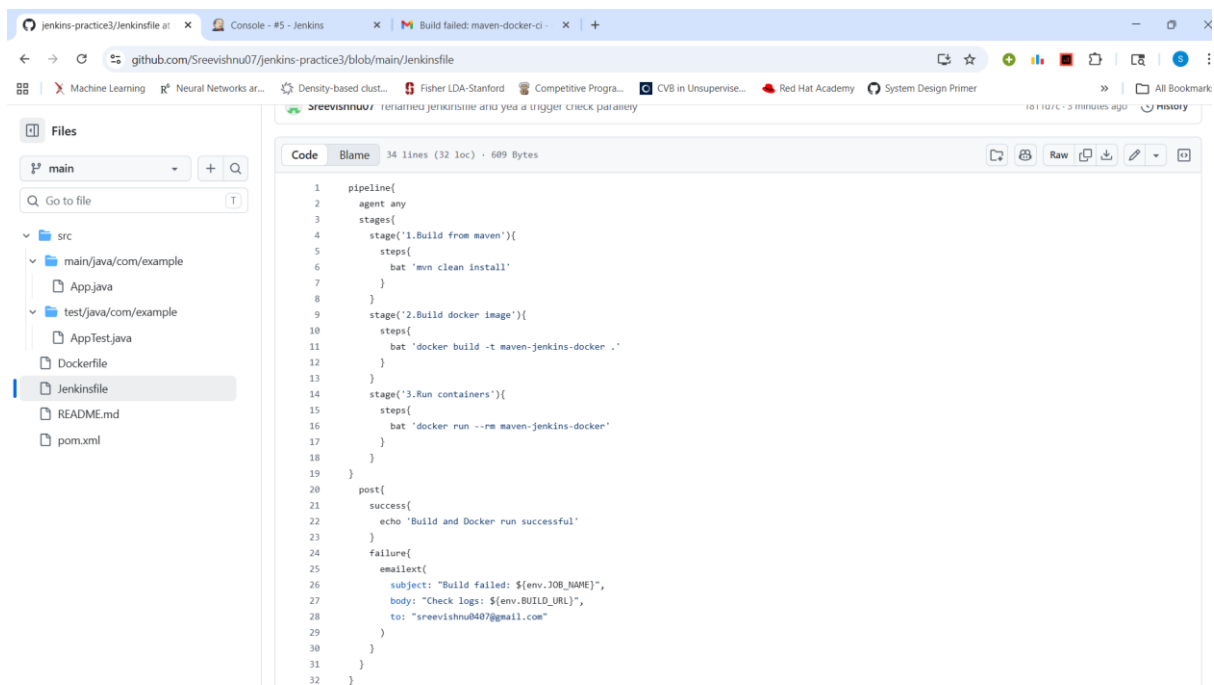
Output from the container run (Dockerfile cmd at play)

Below is the Jenkinsfile based implementation of the above project



The screenshot shows the Jenkins web interface for a build named 'maven-docker-ci' in job '#5'. The console output is displayed, showing the execution of a Jenkinsfile. The output includes the following steps and commands:

```
Started by GitHub push by Sreevishnu07
Obtained Jenkinsfile from git https://github.com/Sreevishnu07/jenkins-practice3
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\jenkins\workspace\maven-docker-ci
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
using credential vishnu
> git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\jenkins\workspace\maven-docker-ci\.git # timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/Sreevishnu07/jenkins-practice3 # timeout=10
Fetching upstream changes from https://github.com/Sreevishnu07/jenkins-practice3
> git.exe --version # timeout=10
> git --version # 'git version 2.43.0.windows.1'
using GIT_ASKPASS to set credentials gmat smtp
> git.exe fetch --tags --force --progress -- https://github.com/Sreevishnu07/jenkins-practice3 +refs/heads/*:refs/remotes/origin/* #
timeout=10
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 1811d7c093c34828953b3b43ba65960d0e2da1a0 (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 1811d7c093c34828953b3b43ba65960d0e2da1a0 # timeout=10
Commit message: "renamed jenkinsfile and yea a trigger check parallelly"
First time build. Skipping changelog.
[Pipeline] }
```



The screenshot shows the GitHub web interface for the repository 'jenkins-practice3'. The 'Jenkinsfile' is selected in the file browser. The code is displayed in a diff view, showing the following content:

```
1 pipeline{
2   agent any
3   stages{
4     stage('1.Build from maven'){
5       steps{
6         bat 'mvn clean install'
7       }
8     }
9     stage('2.Build docker image'){
10      steps{
11        bat 'docker build -t maven-jenkins-docker .'
12      }
13    }
14    stage('3.Run containers'){
15      steps{
16        bat 'docker run --rm maven-jenkins-docker'
17      }
18    }
19  }
20  post{
21    success{
22      echo 'Build and Docker run successful'
23    }
24    failure{
25      emailx{
26        subject: "Build failed: ${env.JOB_NAME}",
27        body: "Check logs: ${env.BUILD_URL}",
28        to: "sreevishnu0407@gmail.com"
29      }
30    }
31  }
32 }
```

```
C:\ProgramData\Jenkins\.jenkins\workspace\maven-docker-ci>docker run --rm maven-jenkins-docker
Sum: 5
Div: 3
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Build and Docker run successful
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```